

Unit 2: Differential Equations - Theoretical Questions (2 Marks Each)

Question 1: Define an exact differential equation and state its characteristic property.

Answer: An exact differential equation is of form $P(x,y)dx + Q(x,y)dy = 0$ where P and Q satisfy $\partial P/\partial y = \partial Q/\partial x$. Its characteristic property is that it represents the total differential of some function $f(x,y) = \text{constant}$.

Question 2: What is a homogeneous differential equation? Give an example.

Answer: A homogeneous differential equation has the form $dy/dx = f(y/x)$ where f is a function of y/x only. Example: $x(dy/dx) = y + x\sqrt{(y^2/x^2)}$ is homogeneous because it can be written as $dy/dx = (y/x) + \sqrt{(y^2/x^2)}$.

Question 3: State the necessary and sufficient condition for the differential equation $P(x,y)dx + Q(x,y)dy = 0$ to be exact.

Answer: The necessary and sufficient condition for $P(x,y)dx + Q(x,y)dy = 0$ to be exact is that $\partial P/\partial y = \partial Q/\partial x$ for all values of x and y in the domain where P and Q are continuous and differentiable.

Question 4: Define an integrating factor for a differential equation.

Answer: An integrating factor is a function $\mu(x,y)$ that, when multiplied with a non-exact differential equation $P(x,y)dx + Q(x,y)dy = 0$, transforms it into an exact equation $\mu P(x,y)dx + \mu Q(x,y)dy = 0$, making it directly integrable.

Question 5: What is a linear differential equation of first order? Write its standard form.

Answer: A linear differential equation of first order has the form $dy/dx + P(x)y = Q(x)$, where $P(x)$ and $Q(x)$ are functions of x only. It is linear in the dependent variable y and its derivatives.

Question 6: Define Bernoulli's differential equation and state how it differs from linear equations.

Answer: Bernoulli's equation has the form $dy/dx + P(x)y = Q(x)y^n$ where $n \neq 0, 1$. It differs from linear equations by including a non-linear term $Q(x)y^n$, requiring substitution $w = y^{1-n}$ to convert it to a linear form.

Question 7: What is a Clairaut's equation? Write its general solution.

Answer: Clairaut's equation has the form $y = xy' + f(y')$ where $y' = dy/dx$. Its general solution is $y = cx + f(c)$ where c is an arbitrary constant, and it also admits a singular solution obtained by differentiating with respect to y' .

Question 8: What is the difference between general and particular solutions of a differential equation?

Answer: A general solution contains arbitrary constants equal to the order of the differential equation and represents all possible solutions. A particular solution is obtained by applying specific initial conditions to determine these constants.

Question 9: Explain the geometric interpretation of a differential equation $dy/dx = f(x,y)$.

Answer: Geometrically, $dy/dx = f(x,y)$ defines a slope field where at each point (x,y) , the value $f(x,y)$ gives the slope of the tangent line to the solution curve passing through that point. Solutions are curves following these tangent directions.

Question 10: Define an orthogonal trajectory and state its application.

Answer: Orthogonal trajectories are curves that intersect every member of a given family of curves at right angles. They are useful in fluid flow problems, electric field analysis, and geometric applications where perpendicular path families need to be determined.

Unit 2: Differential Equations - Numerical Questions (2 Marks Each)

Question 1: Verify whether the differential equation $(x^2 + y^2)dx + 2xydy = 0$ is exact or not.

Solution: For $P(x,y) = x^2 + y^2$ and $Q(x,y) = 2xy$:

- $\partial P/\partial y = 2y$
- $\partial Q/\partial x = 2y$

Since $\partial P/\partial y = \partial Q/\partial x$, the given differential equation is exact.

Question 2: Solve the exact differential equation: $(2xy + y^2)dx + (x^2 + 2xy)dy = 0$

Solution: For $P = 2xy + y^2$ and $Q = x^2 + 2xy$:

- $\partial P/\partial y = 2x + 2y$
- $\partial Q/\partial x = 2x + 2y$

Since $\partial P/\partial y = \partial Q/\partial x$, it's exact. Let $f(x,y) = \text{constant}$, where $\partial f/\partial x = P$ and $\partial f/\partial y = Q$ Integrating $\partial f/\partial x = 2xy + y^2$: $f(x,y) = x^2y + xy^2 + \phi(y)$ Differentiating: $\partial f/\partial y = x^2 + 2xy + \phi'(y) = Q$ Therefore: $\phi'(y) = 0$, $\phi(y) = C_1$ General solution: $x^2y + xy^2 = C$

Question 3: Find the integrating factor for the differential equation $(1 + y^2)dx - xydy = 0$

Solution: Rearranging in the form: $dx/dy = xy/(1+y^2)$

This is a linear equation in x with respect to y : $dx/dy - [y/(1+y^2)]x = 0$

Integrating factor: $\mu = \exp(\int -y/(1+y^2)dy) = \exp(-\frac{1}{2}\ln(1+y^2)) = 1/\sqrt{1+y^2}$

Multiplying original equation: $(1+y^2)dx/\sqrt{1+y^2} - xy/\sqrt{1+y^2}dy = 0$ Solution: $x/\sqrt{1+y^2} = C$ or $x = C\sqrt{1+y^2}$

Question 4: Solve the linear differential equation: $dy/dx + (y/x) = x^2$ where $x > 0$

Solution: This is in the form: $dy/dx + P(x)y = Q(x)$ where $P(x) = 1/x$ and $Q(x) = x^2$

Integrating factor: $\mu = \exp(\int P(x)dx) = \exp(\int dx/x) = \exp(\ln x) = x$

Multiplying both sides: $x(dy/dx) + y = x^3$ $d(xy)/dx = x^3$ $xy = \int x^3 dx = x^4/4 + C$ Solution: $y = x^3/4 + C/x$

Question 5: Solve the Bernoulli's equation: $dy/dx + (y/x) = y^3 x^2$

Solution: This is in form: $dy/dx + P(x)y = Q(x)y^n$ where $P(x) = 1/x$, $Q(x) = x^2$, $n = 3$

Substituting $w = y^{1-n} = y^{-2}$ gives: $dw/dx - 2(y/x)y^{-3} = -2x^2$ $dw/dx - (2/x)w = -2x^2$

Integrating factor: $\mu = \exp(\int -2/x dx) = \exp(-2\ln x) = x^{-2}$ $d(x^{-2}w)/dx = -2x^{-2}w = -2x + C$ $w = -2x^3 + Cx^2$ $y^{-2} = -2x^3 + Cx^2$ Solution: $y = 1/\sqrt{(Cx^2 - 2x^3)}$

Question 6: Find the orthogonal trajectories of the family of curves $xy = c$

Solution: Differentiating $xy = c$: $y + x(dy/dx) = 0$ $dy/dx = -y/x$

Orthogonal trajectory has slope negative reciprocal: $dy/dx = x/y$

This is separable: $y dy = x dx$ Integrating: $y^2/2 = x^2/2 + C$ Solution: $x^2 - y^2 = k$ (family of hyperbolas)

Question 7: Solve the homogeneous differential equation: $(x^2 - y^2)dx + 2xydy = 0$

Solution: Dividing by dx : $(x^2 - y^2) + 2xy(dy/dx) = 0$ Let $y = vx$, then $dy/dx = v + x(dv/dx)$
Substituting: $(x^2 - v^2x^2) + 2x(vx)(v + x(dv/dx)) = 0$ $x^2(1 - v^2) + 2v^2x^2 + 2vx^3(dv/dx) = 0$ $x^2(1 - v^2 + 2v^2) + 2vx^3(dv/dx) = 0$ $x^2(1 + v^2) + 2vx^3(dv/dx) = 0$ $(dv/dx) = -(1 + v^2)/(2vx)$ Separating: $2v/(1 + v^2)dv = -dx/x$ Integrating: $\ln|1 + v^2| = -\ln|x| + \ln|C|$ $(1 + v^2)x = C$ $(1 + y^2/x^2)x = C$ Solution: $x + y^2/x = C$

Question 8: Solve the Clairaut's equation: $y = xy' + (y')^2$

Solution: For Clairaut's equation $y = xp + f(p)$ where $p = y'$, we have: $y = xp + p^2$

General solution: $y = cx + c^2$ where c is constant.

For singular solution, differentiate w.r.t. p : $0 = x + 2p$ $p = -x/2$

Substituting back: $y = x(-x/2) + (-x/2)^2 = -x^2/2 + x^2/4 = -x^2/4$ Singular solution: $y = -x^2/4$

Question 9: Find the particular solution of $dy/dx + 2xy = x$, given $y(0) = 1$

Solution: This is linear with $P(x) = 2x$ and $Q(x) = x$

Integrating factor: $\mu = \exp(\int 2x \, dx) = \exp(x^2) = e^{x^2}$

Multiplying: $e^{x^2}(dy/dx) + 2xe^{x^2}y = xe^{x^2} \Rightarrow d/dx(e^{x^2}y) = xe^{x^2}$
 $e^{x^2}y = \int xe^{x^2} dx = (1/2)e^{x^2} + C$

Therefore: $y = 1/2 + Ce^{-x^2}$ Using $y(0) = 1$: $1 = 1/2 + C$, so $C = 1/2$ Particular solution: $y = 1/2 + (1/2)e^{-x^2}$

Question 10: Determine whether the expression $(2x + 3y^2)dx + (3x + 4y)dy$ is an exact differential. If exact, find its solution.

Solution: For $P = 2x + 3y^2$ and $Q = 3x + 4y$: $\partial P/\partial y = 6y$ $\partial Q/\partial x = 3$

Since $\partial P/\partial y \neq \partial Q/\partial x$, the expression is not exact.

To verify: The original differential equation $(2x + 3y^2)dx + (3x + 4y)dy = 0$ cannot be directly expressed as $d(\text{function}) = 0$, confirming it's not exact.

Unit 3: Ordinary Differential Equations of Higher Order - Theoretical Questions (2 Marks Each)

Question 1: Define a homogeneous linear differential equation of higher order. Give an example.

Answer: A homogeneous linear differential equation of higher order has the form $a_0(x)y^{(n)} + a_1(x)y^{(n-1)} + \dots + a_n(x)y = 0$, where the right-hand side is zero. Example: $x^2y'' - 3xy' + 4y = 0$ is a homogeneous second-order linear differential equation.

Question 2: State the necessary conditions for a second-order linear differential equation to have a solution of the form $y = e^{(rx)}$.

Answer: For a solution $y = e^{(rx)}$ to exist for a second-order linear differential equation with constant coefficients $ay'' + by' + c = 0$, the value r must satisfy the auxiliary equation $ar^2 + br + c = 0$, where a, b, c are coefficients.

Question 3: What is the form of the general solution of a non-homogeneous linear differential equation?

Answer: The general solution of a non-homogeneous linear differential equation is $y = y_c + y_p$, where y_c is the complementary function (general solution of corresponding homogeneous equation) and y_p is a particular integral (any specific solution of non-homogeneous equation).

Question 4: Explain the method of variation of parameters for solving non-homogeneous linear differential equations.

Answer: The method of variation of parameters replaces constants in the complementary function with functions. For $y'' + Py' + Qy = R$, given $y_c = c_1y_1 + c_2y_2$, we find $y_p = u_1y_1 + u_2y_2$ where u_1 and u_2 satisfy specific conditions involving y_1, y_2 and R .

Question 5: What is Cauchy-Euler equation? Write its standard form.

Answer: A Cauchy-Euler equation is a linear differential equation with variable coefficients where each term has the same total degree in x (the independent variable) and derivatives. Standard form: $x^n y^{(n)} + a_1 x^{(n-1)} y^{(n-1)} + \dots + a_n y = f(x)$.

Question 6: Define the Wronskian of a set of functions and state its significance in differential equations.

Answer: The Wronskian $W(f_1, f_2, \dots, f_n)$ is a determinant of matrix containing functions and their derivatives. If $W \neq 0$, the functions are linearly independent and form a fundamental set of solutions for an n th-order homogeneous linear differential equation.

Question 7: When does a second-order linear homogeneous differential equation with constant coefficients have two linearly independent solutions?

Answer: A second-order linear homogeneous differential equation $ay'' + by' + cy = 0$ has two linearly independent solutions when the auxiliary equation $ar^2 + br + c = 0$ has: (i) distinct real roots, (ii) repeated real roots, or (iii) complex conjugate roots.

Question 8: What is the method of undetermined coefficients for solving non-homogeneous linear differential equations?

Answer: Method of undetermined coefficients assumes a particular integral with the same form as non-homogeneous term but with unknown coefficients. These coefficients are determined by substituting the assumed form into the original equation and comparing coefficients.

Question 9: Explain the principle of superposition for linear differential equations.

Answer: The principle of superposition states that if $y_1, y_2, \dots, y_n$ are solutions to a linear homogeneous differential equation, then any linear combination $c_1 y_1 + c_2 y_2 + \dots + c_n y_n$ is also a solution, where $c_1, c_2, \dots, c_n$ are arbitrary constants.

Question 10: What is the significance of the complementary function in solving non-homogeneous differential equations?

Answer: The complementary function represents the general solution of the corresponding homogeneous equation. It contains arbitrary constants equal to the order of the equation and accounts for all possible homogeneous solutions affecting the system's natural response.

Unit 3: Ordinary Differential Equations of Higher Order - Numerical Questions (2 Marks Each)

Question 1: Find the general solution of the differential equation $d^2y/dx^2 - 3dy/dx + 2y = 0$.

Solution: The auxiliary equation is $r^2 - 3r + 2 = 0$ Factorizing: $(r - 1)(r - 2) = 0$ So $r = 1$ or $r = 2$ These are distinct real roots, therefore: General solution: $y = C_1e^x + C_2e^{2x}$

Question 2: Solve the differential equation $d^2y/dx^2 + 4y = 0$.

Solution: The auxiliary equation is $r^2 + 4 = 0$ Solving: $r = \pm 2i$ These are complex conjugate roots, therefore: General solution: $y = C_1\cos(2x) + C_2\sin(2x)$

Question 3: Find the general solution of $d^2y/dx^2 + 6dy/dx + 9y = 0$.

Solution: The auxiliary equation is $r^2 + 6r + 9 = 0$ Factorizing: $(r + 3)^2 = 0$ So $r = -3$ (repeated root) For repeated roots, the general solution is: $y = (C_1 + C_2x)e^{-3x}$

Question 4: Solve the differential equation $d^2y/dx^2 + 25y = 10\sin(5x)$.

Solution: The auxiliary equation is $r^2 + 25 = 0$ Roots: $r = \pm 5i$ Complementary function: $y_c = C_1\cos(5x) + C_2\sin(5x)$

For particular integral, try $y_p = A\cos(5x) + B\sin(5x)$ Substituting and solving: $y_p = -\cos(5x)/5$

General solution: $y = C_1\cos(5x) + C_2\sin(5x) - \cos(5x)/5$

Question 5: Find the general solution of the Cauchy-Euler equation $x^2d^2y/dx^2 - 3xdy/dx + 4y = 0$.

Solution: Substituting $y = x^r$: $r(r-1) - 3r + 4 = 0$ $r^2 - r - 3r + 4 = 0$ $r^2 - 4r + 4 = 0$ $(r - 2)^2 = 0$ So $r = 2$ (repeated root)

General solution: $y = C_1x^2 + C_2x^2\ln(x)$

Question 6: Solve the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^x$.

Solution: Auxiliary equation: $r^2 - 5r + 6 = 0$ Factorizing: $(r - 2)(r - 3) = 0$ Roots: $r = 2, r = 3$
Complementary function: $y_c = C_1e^{2x} + C_2e^{3x}$

For particular integral, try $y_p = Ae^x$ Substituting: $A(1 - 5 + 6)e^x = e^x$ $2Ae^x = e^x$ $A = 1/2$

General solution: $y = C_1e^{2x} + C_2e^{3x} + e^{x/2}$

Question 7: Find the general solution of $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = x^2e^x$.

Solution: Auxiliary equation: $r^2 - 2r + 1 = 0$ Factorizing: $(r - 1)^2 = 0$ Root: $r = 1$ (repeated)
Complementary function: $y_c = (C_1 + C_2x)e^x$

For particular integral, try $y_p = (Ax^2 + Bx + C)xe^x$ Substituting and solving: $y_p = (x^3/6 + x^2/2)e^x$

General solution: $y = (C_1 + C_2x)e^x + (x^3/6 + x^2/2)e^x$

Question 8: Solve the differential equation $\frac{d^2y}{dx^2} - 4y = 8x^2 + 6$.

Solution: Auxiliary equation: $r^2 - 4 = 0$ Roots: $r = \pm 2$ Complementary function: $y_c = C_1e^{2x} + C_2e^{-2x}$

For particular integral, try $y_p = Ax^2 + Bx + C$ Substituting: $2A - 4(Ax^2 + Bx + C) = 8x^2 + 6$ $2A - 4Ax^2 - 4Bx - 4C = 8x^2 + 6$

Comparing coefficients: $-4A = 8$, so $A = -2$ $-4B = 0$, so $B = 0$ $2A - 4C = 6$, so $C = -5/2$

General solution: $y = C_1e^{2x} + C_2e^{-2x} - 2x^2 - 5/2$

Question 9: Find the general solution of $\frac{d^2y}{dx^2} + 9y = \sec(3x)$.

Solution: Auxiliary equation: $r^2 + 9 = 0$ Roots: $r = \pm 3i$ Complementary function: $y_c = C_1\cos(3x) + C_2\sin(3x)$

For particular integral, when forcing term contains $\sec(3x)$, and 3 is a root of auxiliary equation, try: $y_p = xA\cos(3x) + xB\sin(3x)$

Substituting and solving: $y_p = (x/6)\sin(3x)$

General solution: $y = C_1\cos(3x) + C_2\sin(3x) + (x/6)\sin(3x)$

Question 10: Solve the initial value problem $d^2y/dx^2 - 4dy/dx + 4y = 0$, given that $y(0) = 2$ and $y'(0) = 1$.

Solution: Auxiliary equation: $r^2 - 4r + 4 = 0$ Factorizing: $(r - 2)^2 = 0$ Root: $r = 2$ (repeated)

General solution: $y = (C_1 + C_2x)e^{2x}$

Using initial conditions: $y(0) = C_1 = 2$ $y'(0) = 2C_1 + C_2 = 1$

Solving: $C_1 = 2$, $C_2 = -3$

Particular solution: $y = (2 - 3x)e^{2x}$

Unit 4: Partial Differential Equations - Theoretical Questions (2 Marks Each)

Question 1: Define a partial differential equation and explain the order of a PDE.

Answer: A partial differential equation (PDE) is a mathematical equation containing unknown multivariable functions and their partial derivatives. The order of a PDE is determined by the highest-order partial derivative appearing in the equation.

Question 2: What is a homogeneous partial differential equation? Give an example.

Answer: A homogeneous PDE has all terms containing the dependent variable or its derivatives, with no independent terms. Example: $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 0$ (Laplace equation) is homogeneous because every term contains the dependent variable u or its derivatives.

Question 3: Explain the method of separation of variables for solving partial differential equations.

Answer: Separation of variables assumes the solution is a product of functions, each depending on only one variable, i.e., $u(x,y) = X(x)Y(y)$. This transforms the PDE into separate ordinary differential equations, each containing only one independent variable.

Question 4: Define a Cauchy problem for first-order PDE and state the role of characteristic curves.

Answer: A Cauchy problem involves solving a first-order PDE subject to initial conditions specified on a non-characteristic curve. Characteristic curves are paths along which information propagates, and initial data must be prescribed on non-characteristic curves for a well-posed problem.

Question 5: What is a quasi-linear partial differential equation? Give its standard form.

Answer: A quasi-linear PDE is linear in highest-order derivatives but may contain nonlinear terms of lower-order derivatives. For first-order equations, the standard form is $P(x,y,z)p + Q(x,y,z)q = R(x,y,z)$ where $p = \partial z / \partial x$ and $q = \partial z / \partial y$.

Question 6: Define the Lagrange's method for solving first-order quasi-linear PDE.

Answer: Lagrange's method solves PDEs of form $Pp + Qq = R$ by converting to an equivalent system of ODEs $dx/P = dy/Q = dz/R$ called characteristic equations. The solution is obtained by finding two independent integrals of this system.

Question 7: What is the relationship between Lagrange's and Charpit's methods for solving PDEs?

Answer: Lagrange's method solves quasi-linear PDEs $Pp + Qq = R$, while Charpit's method extends this to general first-order PDEs $F(x,y,z,p,q) = 0$. Lagrange's method is a special case of Charpit's method applied when F is linear in p and q .

Question 8: Define a linear partial differential equation and explain its characteristics.

Answer: A linear PDE has the dependent variable and all its derivatives appearing linearly with no products or nonlinear functions. Its characteristics are coefficient functions depend only on independent variables, and the superposition principle applies to its solutions.

Question 9: Explain the concept of particular integrals and complementary functions in PDEs.

Answer: For a non-homogeneous linear PDE, the general solution combines a complementary function (general solution of corresponding homogeneous equation) and a particular integral (any specific solution of non-homogeneous equation), similar to ODEs but with more complexity.

Question 10: What is a well-posed problem in the context of partial differential equations?

Answer: A well-posed problem for a PDE requires existence of a solution, uniqueness of the solution, and continuous dependence of the solution on initial/boundary conditions. If any condition fails, the problem is ill-posed and may not model physical phenomena accurately.

Unit 4: Partial Differential Equations - Numerical Questions (2 Marks Each)

Question 1: Form the partial differential equation by eliminating the arbitrary constants a and b from $z = ax + by + ab$.

Solution: Given $z = ax + by + ab$ Differentiating with respect to x : $\partial z / \partial x = a$ Differentiating with respect to y : $\partial z / \partial y = b$

Let $p = \partial z / \partial x$ and $q = \partial z / \partial y$ Then $a = p$ and $b = q$

Substituting in original equation: $z = px + qy + pq$

The required PDE is: $z = px + qy + pq$ Or equivalently: $z - x\partial z / \partial x - y\partial z / \partial y - (\partial z / \partial x)(\partial z / \partial y) = 0$

Question 2: Form the partial differential equation by eliminating the arbitrary function f from $z = f(x^2 + y^2)$.

Solution: Let $u = x^2 + y^2$, then $z = f(u)$ $\partial z / \partial x = f'(u) \cdot \partial u / \partial x = f'(u) \cdot 2x$ $\partial z / \partial y = f'(u) \cdot \partial u / \partial y = f'(u) \cdot 2y$

Let $p = \partial z / \partial x$ and $q = \partial z / \partial y$ Then $p = 2xf'(u)$ and $q = 2yf'(u)$

Dividing: $p/q = x/y$ This gives: $yp - xq = 0$

The required PDE is: $y\partial z / \partial x - x\partial z / \partial y = 0$

Question 3: Solve the partial differential equation $\partial z / \partial x + \partial z / \partial y = 0$.

Solution: Given: $\partial z / \partial x + \partial z / \partial y = 0$

Using the method of characteristics: $dx/1 = dy/1 = dz/0$

From $dx = dy$, we get $x - y = c_1$ (first integral) From $dz = 0$, we get $z = c_2$ (second integral)

The general solution is: $z = f(x - y)$ where f is an arbitrary function.

Question 4: Find the complete integral of the equation $p^2 + q^2 = 1$, where $p = \partial z / \partial x$ and $q = \partial z / \partial y$.

Solution: Given: $p^2 + q^2 = 1$

Using Charpit's method with complete integral form: $z = ax + by + c$ where $p = a$, $q = b$ and $a^2 + b^2 = 1$

Since $a^2 + b^2 = 1$, we can parameterize: $a = \cos \theta$, $b = \sin \theta$

Substituting: $z = x \cos \theta + y \sin \theta + c$

The complete integral is: $z = x \cos \theta + y \sin \theta + c$ where θ and c are arbitrary parameters.

Question 5: Solve the partial differential equation $(x - y)p + (x + y)q = 0$, where $p = \partial z / \partial x$ and $q = \partial z / \partial y$.

Solution: Given: $(x - y)p + (x + y)q = 0$

Using Lagrange's method: $dx/(x - y) = dy/(x + y) = dz/0$

For $dx/(x - y) = dy/(x + y)$: Integrating: $x^2 - y^2 = c_1$ (first integral)

From $dz = 0$: $z = c_2$ (second integral)

General solution: $z = f(x^2 - y^2)$ where f is an arbitrary function.

Question 6: Find the general solution of the partial differential equation $xp - yq = 0$, where $p = \partial z / \partial x$ and $q = \partial z / \partial y$.

Solution: Given: $xp - yq = 0$ or $x\partial z / \partial x - y\partial z / \partial y = 0$

Using Lagrange's method: $dx/x = dy/(-y) = dz/0$

From $dx/x = dy/(-y)$: $\ln|x| + \ln|y| = \ln|c_1|$ $xy = c_1$ (first integral)

From $dz = 0$: $z = c_2$ (second integral)

The general solution is: $z = f(xy)$ where f is an arbitrary function.

Question 7: Solve the linear PDE $\partial z / \partial x + 2\partial z / \partial y = x + 2y$.

Solution: Given: $\partial z/\partial x + 2\partial z/\partial y = x + 2y$

Using the method of characteristics: $dx/1 = dy/2 = dz/(x + 2y)$

From $dx/1 = dy/2$: $dy = 2dx$ $y = 2x + c_1$

Substituting to find z : $dz = (x + 2y)dx = (x + 2(2x + c_1))dx = (x + 4x + 2c_1)dx = (5x + 2c_1)dx$

Integrating: $z = 5x^2/2 + 2c_1x + c_2$

Substituting back $c_1 = y - 2x$: $z = 5x^2/2 + 2(y - 2x)x + c_2 = 5x^2/2 + 2xy - 4x^2 + c_2 = 2xy - 3x^2/2 + \phi(y - 2x)$

The general solution is: $z = 2xy - 3x^2/2 + \phi(y - 2x)$ where ϕ is an arbitrary function.

Question 8: Form the partial differential equation by eliminating the arbitrary functions f and g from $z = f(x + y) + g(x - y)$.

Solution: Let $u = x + y$, $v = x - y$, then $z = f(u) + g(v)$

$\partial z/\partial x = f'(u) \cdot \partial u/\partial x + g'(v) \cdot \partial v/\partial x = f'(u) + g'(v)$
 $\partial z/\partial y = f'(u) \cdot \partial u/\partial y + g'(v) \cdot \partial v/\partial y = f'(u) - g'(v)$

Therefore: $\partial z/\partial x + \partial z/\partial y = 2f'(u)$ $\partial z/\partial x - \partial z/\partial y = 2g'(v)$

Differentiating again: $\partial^2 z/\partial x^2 + 2\partial^2 z/\partial x\partial y + \partial^2 z/\partial y^2 = 2f''(u)$ $\partial^2 z/\partial x^2 - 2\partial^2 z/\partial x\partial y + \partial^2 z/\partial y^2 = 2g''(v)$

Eliminating f'' and g'' : $\partial^2 z/\partial x\partial y = 0$

The required PDE is: $\partial^2 z/\partial x\partial y = 0$

Question 9: Solve the PDE $x\partial z/\partial x + y\partial z/\partial y = z$ using the method of characteristics.

Solution: Given: $x\partial z/\partial x + y\partial z/\partial y = z$ or $xp + yq = z$

Using Lagrange's method: $dx/x = dy/y = dz/z$

From $dx/x = dy/y$: $\ln|x| = \ln|y| + \ln|c_1|$ $y/x = c_1$ (first integral)

From $dx/x = dz/z$: $\ln|x| = \ln|z| + \ln|c_2|$ $z/x = c_2$ (second integral)

The general solution is: $z = x\phi(y/x)$ where ϕ is an arbitrary function.

Question 10: Find the complete integral of the equation $p^2 - q^2 = x^2 - y^2$, where $p = \partial z / \partial x$ and $q = \partial z / \partial y$.

Solution: Given: $p^2 - q^2 = x^2 - y^2$

Using Charpit's method with complete integral form: $z = ax + by + c$ where $p = a$, $q = b$

Substituting: $a^2 - b^2 = x^2 - y^2$

This equation must be valid for all x and y , which means: $a^2 = x^2 + \alpha$ and $b^2 = y^2 + \alpha$, where α is a parameter

So $a = \pm\sqrt{x^2 + \alpha}$ and $b = \pm\sqrt{y^2 + \alpha}$

For simplicity, take positive values: $z = x\sqrt{x^2 + \alpha} + y\sqrt{y^2 + \alpha} - \int(x^2 + \alpha)^{1/2}dx - \int(y^2 + \alpha)^{1/2}dy + c$

Evaluating integrals: $z = x\sqrt{x^2 + \alpha} + y\sqrt{y^2 + \alpha} - (x^2 + \alpha)^{3/2}/(3x) - (y^2 + \alpha)^{3/2}/(3y) + c$

Simplifying: $z = x\sqrt{x^2 + \alpha} + y\sqrt{y^2 + \alpha} - 2(x^2 + \alpha)^{3/2}/3 - 2(y^2 + \alpha)^{3/2}/3 + c$

The complete integral is: $z = x\sqrt{x^2 + \alpha} + y\sqrt{y^2 + \alpha} - 2(x^2 + \alpha)^{3/2}/3 - 2(y^2 + \alpha)^{3/2}/3 + c$ where α and c are arbitrary parameters.

Unit 5: Higher Order Partial Differential Equations - Theoretical Questions (2 Marks Each)

Question 1: Define the wave equation and explain its physical significance.

Answer: The wave equation $\partial^2 u / \partial t^2 = c^2 \nabla^2 u$ describes the propagation of waves through a medium, where c is wave speed. It models vibrations, sound waves, electromagnetic waves, and other oscillatory phenomena, representing the balance between acceleration and restoring forces in wave motion.

Question 2: What is the heat equation and what physical process does it model?

Answer: The heat equation $\partial u / \partial t = \alpha \nabla^2 u$ describes temperature distribution in a material over time, where α is thermal diffusivity. It models heat conduction, diffusion processes, and other phenomena where a quantity spreads from regions of high concentration to low concentration.

Question 3: Define the Laplace equation and state its significance in mathematical physics.

Answer: The Laplace equation $\nabla^2 u = 0$ describes steady-state or equilibrium problems with no time dependence. It models potential fields in electrostatics, gravitational fields, steady-state temperature distributions, and fluid flows where quantities have reached equilibrium configurations.

Question 4: Explain the D'Alembert's solution for the one-dimensional wave equation.

Answer: D'Alembert's solution expresses the solution to the one-dimensional wave equation as $u(x,t) = f(x+ct) + g(x-ct)$, representing superposition of two waves traveling in opposite directions with speed c , where f and g are determined by initial conditions.

Question 5: What are boundary conditions in PDEs? Explain Dirichlet and Neumann boundary conditions.

Answer: Boundary conditions specify constraints on solutions at domain boundaries. Dirichlet conditions prescribe the function value ($u = f$) at boundaries, while Neumann conditions prescribe the normal derivative ($\partial u / \partial n = g$), representing flux across boundaries.

Question 6: Define homogeneous and non-homogeneous PDEs and give examples of each.

Answer: Homogeneous PDEs have zero on the right side (e.g., $\nabla^2 u = 0$, Laplace equation). Non-homogeneous PDEs have non-zero terms on the right side (e.g., $\nabla^2 u = f(x,y,z)$, Poisson equation), representing source terms, forcing functions, or external influences in physical systems.

Question 7: Explain the classification of second-order PDEs into elliptic, parabolic, and hyperbolic types.

Answer: Second-order PDEs are classified by discriminant $B^2 - 4AC$ for general form $Au_{xx} + Bu_{xy} + Cu_{yy} + \dots = 0$. Elliptic ($B^2 - 4AC < 0$) model equilibrium phenomena (Laplace). Parabolic ($B^2 - 4AC = 0$) model diffusion (heat equation). Hyperbolic ($B^2 - 4AC > 0$) model wave propagation (wave equation).

Question 8: What is a Green's function and how is it used to solve PDEs?

Answer: A Green's function $G(x, \xi)$ is the response to a point source in a differential equation. For linear PDEs, the solution is obtained by convolving the Green's function with source terms, allowing complex problems to be solved through integration of simpler point-source solutions.

Question 9: Explain the principle of superposition as applied to linear partial differential equations.

Answer: The principle of superposition states that for linear PDEs, if u_1 and u_2 are solutions, then any linear combination $c_1 u_1 + c_2 u_2$ is also a solution. This allows complex problems to be decomposed into simpler ones whose solutions can be combined.

Question 10: What are characteristic curves for second-order PDEs and how do they determine the type of PDE?

Answer: Characteristic curves are paths along which information propagates in PDEs. For second-order PDEs, their existence and nature determine the classification: hyperbolic PDEs have two real characteristic directions, parabolic have one repeated direction, and elliptic have no real characteristics.

Unit 5: Higher Order Partial Differential Equations - Numerical Questions (2 Marks Each)

Question 1: Classify the following partial differential equation and find its canonical form: $3\partial^2 z/\partial x^2 + 4\partial^2 z/\partial x\partial y + \partial^2 z/\partial y^2 = 0$

Solution: For the equation $3\partial^2 z/\partial x^2 + 4\partial^2 z/\partial x\partial y + \partial^2 z/\partial y^2 = 0$: $A = 3$, $B = 4$, $C = 1$

$$\text{Discriminant} = B^2 - 4AC = 4^2 - 4(3)(1) = 16 - 12 = 4 > 0$$

Since the discriminant is positive, the equation is hyperbolic.

For canonical form, find characteristic directions: $B^2 - 4AC = 4 = (2)^2$ Therefore, characteristic curves have slopes: $m_1 = (-B + \sqrt{B^2 - 4AC})/(2A) = (-4 + 2)/(6) = -1/3$ $m_2 = (-B - \sqrt{B^2 - 4AC})/(2A) = (-4 - 2)/(6) = -1$

Using $\xi = x - y/3$, $\eta = x + y$ as new variables: The canonical form is: $\partial^2 z/\partial \xi \partial \eta = 0$

Question 2: Solve the wave equation $\partial^2 u/\partial t^2 = 9\partial^2 u/\partial x^2$ subject to the conditions $u(x,0) = \sin(2x)$ and $\partial u/\partial t(x,0) = 0$.

Solution: Given: $\partial^2 u/\partial t^2 = 9\partial^2 u/\partial x^2$ with $u(x,0) = \sin(2x)$ and $u_t(x,0) = 0$ Wave speed $c = 3$

Using D'Alembert's solution: $u(x,t) = (1/2)[f(x+ct) + f(x-ct)] + (1/2c)\int_{x-ct}^{x+ct} g(s)ds$

Here $f(x) = \sin(2x)$ and $g(x) = 0$

$$u(x,t) = (1/2)[\sin(2(x+3t)) + \sin(2(x-3t))] = (1/2)[\sin(2x+6t) + \sin(2x-6t)] = \sin(2x)\cos(6t)$$

Therefore: $u(x,t) = \sin(2x)\cos(6t)$

Question 3: Solve the heat equation $\partial u/\partial t = \partial^2 u/\partial x^2$ subject to $u(0,t) = 0$, $u(L,t) = 0$, $u(x,0) = \sin(\pi x/L)$.

Solution: Using separation of variables, let $u(x,t) = X(x)T(t)$

This gives: $X''(x)/X(x) = T'(t)/T(t) = -\lambda^2$

For boundary conditions $X(0) = X(L) = 0$: $X(x) = \sin(n\pi x/L)$ where n is a positive integer

The eigenvalues are $\lambda_n = n\pi/L$

The time function satisfies: $T'(t) = -(n\pi/L)^2 T(t)$ $T(t) = e^{-\{(n\pi/L)^2 t\}}$

Initial condition $u(x,0) = \sin(\pi x/L)$ matches exactly $n = 1$ Therefore, $u(x,t) = \sin(\pi x/L)e^{-\{(\pi/L)^2 t\}}$

Question 4: Solve Laplace's equation $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 0$ in a rectangular region $0 \leq x \leq a$, $0 \leq y \leq b$ with boundary conditions $u(0,y) = 0$, $u(a,y) = 0$, $u(x,0) = 0$, $u(x,b) = f(x)$.

Solution: Using separation of variables, let $u(x,y) = X(x)Y(y)$

This gives: $X''(x)/X(x) = -Y''(y)/Y(y) = -\lambda^2$

For $X(0) = X(a) = 0$: $X(x) = \sin(n\pi x/a)$ where n is a positive integer

For Y : $Y''(y) - \lambda^2 Y(y) = 0$ $Y(y) = A_n \sinh(n\pi y/a) + B_n \cosh(n\pi y/a)$

From $u(x,0) = 0$: $Y(0) = 0$ implies $B_n = 0$

Therefore: $u(x,y) = \sum_n C_n \sin(n\pi x/a) \sinh(n\pi y/a)$

From $u(x,b) = f(x)$: $f(x) = \sum_n C_n \sinh(n\pi b/a) \sin(n\pi x/a)$

Thus: $C_n = (2/a) \cdot (1/\sinh(n\pi b/a)) \cdot \int_0^a f(x) \sin(n\pi x/a) dx$

Therefore: $u(x,y) = \sum_n [(2/a) \cdot \int_0^a f(x) \sin(n\pi x/a) dx] \cdot [\sinh(n\pi y/a) / \sinh(n\pi b/a)] \cdot \sin(n\pi x/a)$

Question 5: Classify the following partial differential equation: $x^2 \partial^2 z / \partial x^2 - y^2 \partial^2 z / \partial y^2 = 0$

Solution: Rewriting the equation in standard form: $x^2 \partial^2 z / \partial x^2 - y^2 \partial^2 z / \partial y^2 = 0$

Here $A = x^2$ (coefficient of $\partial^2 z / \partial x^2$) $B = 0$ (coefficient of $\partial^2 z / \partial x \partial y$) $C = -y^2$ (coefficient of $\partial^2 z / \partial y^2$)

Discriminant $= B^2 - 4AC = 0^2 - 4(x^2)(-y^2) = 4x^2 y^2 > 0$ (for $x \neq 0$, $y \neq 0$)

Since the discriminant is positive, the equation is hyperbolic in regions where $x \neq 0$ and $y \neq 0$.

Question 6: Solve the wave equation $\partial^2 u / \partial t^2 = 4 \partial^2 u / \partial x^2$ for $0 < x < \pi$ with boundary conditions $u(0, t) = 0$, $u(\pi, t) = 0$ and initial conditions $u(x, 0) = \sin(3x)$, $\partial u / \partial t(x, 0) = 0$.

Solution: Using separation of variables, let $u(x, t) = X(x)T(t)$

For the boundary conditions $X(0) = X(\pi) = 0$: $X(x) = \sin(nx)$ where n is a positive integer

The eigenvalues are $\lambda_n = n^2$

The time function satisfies: $T''(t) + 4n^2 T(t) = 0$ $T(t) = A_n \cos(2nt) + B_n \sin(2nt)$

From initial condition $\partial u / \partial t(x, 0) = 0$: $B_n = 0$

Therefore: $u(x, t) = \sum_n A_n \sin(nx) \cos(2nt)$

From initial condition $u(x, 0) = \sin(3x)$: $\sin(3x) = \sum_n A_n \sin(nx)$

By orthogonality of sine functions: $A_n = 0$ for $n \neq 3$ and $A_3 = 1$

Therefore: $u(x, t) = \sin(3x) \cos(6t)$

Question 7: Find the solution of the heat equation $\partial u / \partial t = \partial^2 u / \partial x^2$ for $0 < x < 1$, $t > 0$ with boundary conditions $u(0, t) = 0$, $u(1, t) = 0$ and initial condition $u(x, 0) = x(1-x)$.

Solution: Using separation of variables, let $u(x, t) = X(x)T(t)$

For boundary conditions $X(0) = X(1) = 0$: $X(x) = \sin(n\pi x)$ where n is a positive integer

The eigenvalues are $\lambda_n = n^2 \pi^2$

The time function satisfies: $T'(t) = -n^2 \pi^2 T(t)$ $T(t) = e^{-n^2 \pi^2 t}$

Therefore: $u(x, t) = \sum_n A_n \sin(n\pi x) e^{-n^2 \pi^2 t}$

From initial condition $u(x, 0) = x(1-x)$: $x(1-x) = \sum_n A_n \sin(n\pi x)$

Using Fourier sine series: $A_n = 2 \int_0^1 x(1-x) \sin(n\pi x) dx$

Integrating by parts: $A_n = 2[2(1 - \cos(n\pi)) / (n^3 \pi^3)]$

Since $1 - \cos(n\pi) = 0$ for even n and 2 for odd n : $A_n = 0$ for even n $A_n = 8 / (n^3 \pi^3)$ for odd n

Therefore: $u(x,t) = \sum_{n \text{ odd}} [8/(n^3\pi^3)] \sin(n\pi x) e^{-n^2\pi^2 t}$

Question 8: Solve Laplace's equation $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 0$ in a semi-infinite strip $0 < x < a$, $y > 0$ with boundary conditions $u(0,y) = 0$, $u(a,y) = 0$, $u(x,0) = f(x)$.

Solution: Using separation of variables, let $u(x,y) = X(x)Y(y)$

For boundary conditions $X(0) = X(a) = 0$: $X(x) = \sin(n\pi x/a)$ where n is a positive integer

For Y , we need solutions that decay as $y \rightarrow \infty$: $Y''(y) - (n\pi/a)^2 Y(y) = 0$ $Y(y) = C_n e^{-(n\pi/a)y}$

Therefore: $u(x,y) = \sum_n B_n \sin(n\pi x/a) e^{-(n\pi/a)y}$

From initial condition $u(x,0) = f(x)$: $f(x) = \sum_n B_n \sin(n\pi x/a)$

Using Fourier sine series: $B_n = (2/a) \int_0^a f(x) \sin(n\pi x/a) dx$

Therefore: $u(x,y) = \sum_n [(2/a) \int_0^a f(x) \sin(n\pi x/a) dx] \sin(n\pi x/a) e^{-(n\pi/a)y}$

Question 9: Transform the equation $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 - 2\partial^2 u / \partial x \partial y = 0$ to canonical form and classify it.

Solution: For the equation $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 - 2\partial^2 u / \partial x \partial y = 0$: $A = 1$, $B = -2$, $C = 1$

Discriminant = $B^2 - 4AC = (-2)^2 - 4(1)(1) = 4 - 4 = 0$

Since the discriminant is zero, the equation is parabolic.

For canonical form, find the characteristic direction: $m = (-B \pm \sqrt{B^2 - 4AC}) / (2A) = -(-2) \pm \sqrt{0} / (2) = 1$

Using $\xi = x + y$, $\eta = y$ as new variables: The canonical form is: $\partial^2 u / \partial \eta^2 = 0$

Question 10: Solve the Poisson equation $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 4$ in the square region $0 < x < 1$, $0 < y < 1$ with boundary conditions $u = 0$ on all sides.

Solution: The solution consists of complementary function u_c (solution of Laplace equation) and particular integral u_p (solution of Poisson equation): $u = u_c + u_p$

For u_p , try $u_p = Ax^2 + By^2$ Substituting into $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 4$: $2A + 2B = 4$, giving $A = B = 1$

Therefore $u_p = x^2 + y^2$

For u_c , using separation of variables: $u_c = \sum_m \sum_n A_{mn} \sin(m\pi x) \sin(n\pi y) e^{-\pi \sqrt{m^2 + n^2} y}$

Applying boundary condition $u = 0$ at $y = 0$: $u_p = x^2 + y^2 - 0 = x^2 + y^2$

Applying boundary condition $u = 0$ at $y = 1$: $x^2 + 1 + u_c(x, 1) = 0$ $u_c(x, 1) = -(x^2 + 1)$

This gives: $A_{mn} = -(4/(\pi^2 mn)) [1 - (-1)^m] [1 - (-1)^n]$

For boundary conditions at $x = 0$ and $x = 1$: Similar expressions are found.

The complete solution: $u(x, y) = x^2 + y^2 + \sum_m \sum_n A_{mn} \sin(m\pi x) \sin(n\pi y)$

where coefficients A_{mn} are determined by Fourier series of $-(x^2 + y^2)$ on the boundary.